

BDS-SMC2 Final Hardware Day — Field Sheet

Sheet 1 — Morning Setup: Portal Activation + 112-bit Firmware

Date	Start time	Location (same as Gap 2 baseline)	Experimenter
			Letsoalo Maile

[]	A. Portal reader activation (at home, before leaving)	Done time
[]	Log in at http://bdrd.hwasmart.com (RCSSTEAP_3058_SM_1 / 123456)	
[]	F12 → Application → Local Storage → copy data , access_token , refresh_token	
[]	Paste into <code>python/portal_config.json</code>	
[]	Run: <code>python python/portal_reader.py --dump</code> (record real message field names below)	
[]	Start dashboard: <code>python python/tx_dashboard.py</code> → http://localhost:8765	

Portal message field names seen in --dump output:

Field name	Example value	Maps to

[]	B. Firmware flash (112-bit payload) — disconnect GPIO16/17 first, DIO mode, 40 MHz, hold BOOT + press RESET at 'Connecting...'	Done
[]	Flash <code>esp32_sender.ino</code> (MODE=1 binary; test coord = lab T001)	
[]	Reconnect GPIO16/17; power on; wait 2 minutes before first TX	
[]	Verify serial shows: [BINARY TX] \$CCTXM,0,BIN:1D35DB5605079637007200A00101*..	

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Sheet — Gap 2 Morning Session · 30 TX · MODE 1 (112-bit binary) · run 08:00–10:00

Session	Start (HH:MM)	Weather	Cloud %	Temp (opt.)
morning				

Command: `.\run_gap2_morning.bat` (auto-stops at 30 TX). Power on ESP32 → wait 2 min → run bat → power off immediately after.

Option B: ALL sessions transmit the 112-bit binary payload (one flash, no mode switching). Logger fills `gap2_latency.csv` automatically — use this sheet only for anomalies.

TX block	All OK? (Y/N)	Anomalies (TX#, what happened)
1–5		
6–10		
11–15		
16–20		
21–25		
26–30		

Metric	Value
Successes / 30	
Timeouts	
Mean latency from logger (ms)	
Dashboard 'ground confirmed' count	

112-bit acceptance check (first TX of this session IS the hardware verification):

Check	Result
TX #1: module accepted 28-char hex (no \$CCTXM error)? ☐ yes ☐ no	
TX #1: [T3] Send Success + green LED? ☐ yes ☐ no	
TX #1 visible on portal (screenshot taken)? ☐ yes ☐ no	
CRITICAL — portal shows: ☐ packed binary (112 bits on-air) ☐ ASCII hex text (224 bits)	
IF MODULE REJECTS: stop session — fallback firmware (R as uint8, 104 bits) needed first	

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Sheet — Gap 2 Midday Session · 30 TX · MODE 1 (112-bit binary) · run 12:00–14:00 — SAME DAY

Session	Start (HH:MM)	Weather	Cloud %	Temp (opt.)
midday				

Command: `.\run_gap2_midday.bat` (auto-stops at 30 TX). Power on ESP32 → wait 2 min → run bat → power off immediately after.

Option B: ALL sessions transmit the 112-bit binary payload (one flash, no mode switching). Logger fills `gap2_latency.csv` automatically — use this sheet only for anomalies.

TX block	All OK? (Y/N)	Anomalies (TX#, what happened)
1–5		
6–10		
11–15		
16–20		
21–25		
26–30		

Metric	Value
Successes / 30	
Timeouts	
Mean latency from logger (ms)	
Dashboard 'ground confirmed' count	

Optional — power measurement (USB meter in supply line, during this session):

State	Voltage (V)	Current (mA)	Duration (s)	Notes
Idle (module on, no TX)				
During TX burst				
Peak observed				

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Sheet — Gap 2 Evening Session · 30 TX · MODE 1 (112-bit binary) · run AFTER 18:00 — SAME DAY

Session	Start (HH:MM)	Weather	Cloud %	Temp (opt.)
evening				

Command: `.\run_gap2_evening.bat` (auto-stops at 30 TX). Power on ESP32 → wait 2 min → run bat → power off immediately after.

Option B: ALL sessions transmit the 112-bit binary payload (one flash, no mode switching). Logger fills `gap2_latency.csv` automatically — use this sheet only for anomalies.

TX block	All OK? (Y/N)	Anomalies (TX#, what happened)
1–5		
6–10		
11–15		
16–20		
21–25		
26–30		

Metric	Value
Successes / 30	
Timeouts	
Mean latency from logger (ms)	
Dashboard 'ground confirmed' count	

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